Chapter 8. Conditional Independence and Gaussian Graphical Models

Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighborhood selection)

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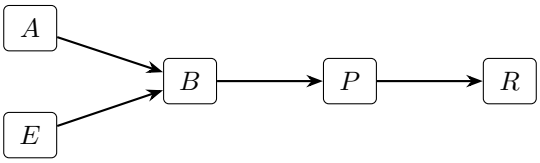
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Roadmap of the causal-discovery part

Lecture	Condition on the data	Relation to be discovered
1 (Ch. 8)	Gaussian observational data	undirected graph of conditional relations
2 (Ch. 9)	linear SEM with given directions	conditional independences a graph implies; limits of direction
3 (Ch. 10)	all main variables observed	the equivalence class of DAGs that data can distinguish
4 (Ch. 11)	hidden common causes allowed	latent confounding, relations and directions compatible with it
5 (Ch. 12)	extra assumption on the error distribution	more specific directions inside the equivalence class
6 (Ch. 13)	further identifiable models	directions for which causal discovery is possible

A running toy health example

Used in all six lectures:



A	age	E	amount of exercise
B	body-mass index (BMI)	P	blood pressure
R	disease risk		

Lecture 4 adds a hidden lifestyle variable U; Lecture 6 adds measurement error.

1 Correlation Mixes Direct and Indirect Relations

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ independent, so all variances equal 1.

Pair	Correlation	Why the relation arises
X_1, X_2	0.8	X_1 and X_2 are directly connected
X_2, X_3	0.8	X_2 and X_3 are directly connected
X_1, X_3	0.64	the relation is transmitted through X_2

$\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The question is: **does a relation remain after accounting for the other variables?**

2 Conditional Independence

Definition: conditional independence

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all z, equivalently $p(x \mid y, z) = p(x \mid z)$.

In the chain, $X_1 \not\perp\!\!\!\perp X_3$ (Corr $= 0.64$) but $X_1 \perp\!\!\!\perp X_3 \mid X_2$:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix direction, but it **narrows the candidate edges**.

3 Partial Correlation

Motivation. Partial correlation asks whether a direct association remains once the other variables are removed linearly.

3.1 Definition by residuals

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

Quantity	Meaning
r_1	the part of X_1 not explained by X_2
r_3	the part of X_3 not explained by X_2
$\mathrm{Corr}(r_1, r_3)$	the partial correlation of X_1, X_3 given X_2

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals; $\rho_{jk\cdot S}$ regresses both on X_S.

Theorem: first-order partial correlation

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Proof [Supplement]. Standardize. Then $r_1 = X_1 - \rho_{12}X_2$, $r_3 = X_3 - \rho_{23}X_2$, so $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$, $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

Example: a correlation of 0.64 can disappear

With $\rho_{12} = \rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0,$$

so under Gaussianity $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

4 The Precision Matrix

4.1 Zeros of the precision matrix are conditional independences

Let $X \sim N_p(\mu, \Sigma)$ and $\Omega = \Sigma^{-1}$ (the **precision matrix**).

Precision matrix	Interpretation
$\Omega_{jk} = 0$	$X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$
$\Omega_{jk} \neq 0$	a relation between X_j, X_k remains given all the other variables

Theorem: precision matrix and full-order partial correlation

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

Proof [Supplement]. Let $A = \{j, k\}$, B the rest. $\mathrm{Cov}(X_A \mid X_B) = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = (\Omega_{AA})^{-1}$ (block inversion), so

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

with correlation $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$, free of x_B, hence the partial correlation. For a bivariate Gaussian, zero correlation $\iff$ independence. ■

4.2 Turning the precision matrix into a graph: the GGM

Definition: Gaussian graphical model

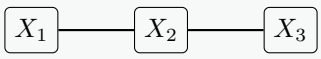
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \ldots, p\}$ with

$$(j, k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

Example: the three-variable chain

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$. GGM:



$\rho_{12\cdot 3} = 0.8/\sqrt{1.64} \approx 0.62 < 0.8$.

Remark: from pairwise to global statements

For a positive density the pairwise Markov property is equivalent to the global one: if S separates A and B in the graph, $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford).

5 Estimating the Graph

5.1 Inverting the sample covariance produces spurious edges

The naive estimate is $\widehat{\Omega} = S^{-1}$.

Data condition	Problem
finite sample	because of sampling error, entries that are truly 0 are estimated as small nonzero values
many variables	S^{-1} is unstable and most entries are nonzero
$p \geq n$	S is singular and the inverse may not exist

We need a **sparse** estimate.

Method	Core idea	Suitable when
partial-correlation tests	compute partial correlations from S^{-1} and test $H_0 : \rho_{jk\cdot -\{j,k\}} = 0$	$p < n$ and sample large enough
graphical lasso	ℓ_1 penalty on the whole precision matrix, some entries estimated as 0	a sparse positive-definite precision matrix is needed
neighborhood selection	Lasso regression of each variable on the rest; choose neighbors variable by variable	many variables, node-wise relations
CLIME	directly estimate a sparse Ω satisfying $S\Omega \approx I$	estimating the precision matrix in high dimension

5.2 Testing a zero partial correlation

For Gaussian data, $X_j \perp\!\!\!\perp X_k \mid X_S \iff \rho_{jk\cdot S} = 0$. Let $\hat\rho = \hat\rho_{jk\cdot S}$, $q = |S|$.

Test	Statistic	Reference distribution
partial-correlation t-test	$T = \hat\rho\sqrt{\dfrac{n-q-2}{1-\hat\rho^2}}$	t_{n-q-2}
Fisher-z test	$Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat\rho)$	approximately $N(0,1)$
conditional distance correlation	$\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$	p-value by resampling

Derivation: where the degrees of freedom come from

Regressing on q variables plus an intercept leaves a residual correlation like one from $m = n - q$ pairs. For m pairs and $\rho = 0$, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$, and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method ($\mathrm{Var}(r) \approx (1-\rho^2)^2/m$, $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$).

Failing to reject H_0 does *not* prove conditional independence.

5.3 Graphical lasso

Derivation: Gaussian log-likelihood of Ω

For centered data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top\Omega x_i + c$ and $\sum_i x_i^\top\Omega x_i = n\,\mathrm{tr}(S\Omega)$, so the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \mathrm{tr}(S\Omega)\} + c$, maximized at S^{-1} when $S \succ 0$.

$$\widehat\Omega_\lambda = \arg\min_{\Omega\succ 0}\Big[\mathrm{tr}(S\Omega) - \log\det(\Omega) + \lambda\sum_{j\neq k}|\Omega_{jk}|\Big].$$

Part of the objective	Role
$\mathrm{tr}(S\Omega) - \log\det(\Omega)$	matches the covariance structure of the observed data
$\lambda\sum_{j\neq k}\lvert\Omega_{jk}\rvert$	sets small off-diagonal entries exactly to 0

Convex; optimality: $\Omega^{-1} - S - \lambda\Gamma = 0$, $\Gamma_{jk} \in \partial|\Omega_{jk}|$, $\Gamma_{jj} = 0$. Hence $|(\widehat\Omega^{-1})_{jk} - S_{jk}| \le \lambda$, with equality on edges. The solution is positive definite even for $p \ge n$; λ by CV log-likelihood, EBIC or stability selection.

5.4 Neighborhood selection

Theorem: conditional distribution of one node

For centered $X \sim N_p(0, \Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}} X_{-j},\ \frac{1}{\Omega_{jj}} \Big),$$

so the population coefficients of the regression of X_j on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

Proof [Supplement]. Schur complement with $A = \{j\}$: $\mathrm{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1} X_{-j} = -\Omega_{jj}^{-1}\Omega_{j,-j} X_{-j}$ by $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

Estimate by the Lasso:

$$\hat\beta^{(j)} = \arg\min_{\beta} \Big\{ \frac{1}{2n} \|X_j - X_{-j}\beta\|_2^2 + \lambda \|\beta\|_1 \Big\}.$$

$\hat\beta_k^{(j)} \neq 0$ makes $j-k$ a candidate; the two regressions are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one). It parallelises over nodes but gives a graph, not a positive-definite $\widehat\Omega$.

5.5 CLIME

$$\widehat\Omega = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \le \lambda,$$

a column-wise linear program, symmetrized afterwards.

6 GGM Edges Are Conditional Associations, Not Causal Directions

Interpretable from a GGM	Not interpretable from a GGM alone
Does a relation remain after accounting for the other observed variables?	Which variable is the cause and which the effect?
What are the sign and size of the partial correlation?	Is an edge a direct causal effect?
Which variables are conditionally connected to many others?	Is there a hidden common cause?

$X_j - X_k$ means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$; conditioning on a common *effect* can even create an edge between independent causes (Ch. 9). Directions need DAGs.

Summary

- Partial correlations are residual correlations; for Gaussians $\rho_{jk\cdot\mathrm{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$, so $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$.
- GGM edge $\iff$ $\Omega_{jk} \neq 0$; estimated by tests, graphical lasso, neighborhood selection or CLIME.
- GGM edges are undirected conditional associations.

7 Selected exercises

Motivation. These prove the block-inverse identity and Gaussian conditioning, the graphical-lasso $\lambda_{\max}$, and the link between partial correlations and node-wise regressions.

Exercise 8.1 (Block inverse and Gaussian conditioning)

Let $\Sigma \succ 0$ be partitioned into blocks A, B, $\Omega = \Sigma^{-1}$, and $M = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$.
(a) Show that $M \succ 0$, $\Omega_{AA} = M^{-1}$ and $\Omega_{AB} = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.
(b) For $X \sim N_p(\mu, \Sigma)$ show that $X_A \mid X_B = x_B \sim N(\mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B),\, M)$.

Solution [Supplement].

(a)

Step 1. For $v \neq 0$, $u = (v^\top, -(\Sigma_{BB}^{-1}\Sigma_{BA}v)^\top)^\top \neq 0$ gives $v^\top M v = u^\top \Sigma u > 0$, so $M \succ 0$.

Step 2. With $K = \Sigma_{AB}\Sigma_{BB}^{-1}$ let

$$\Omega' = \begin{pmatrix} M^{-1} & -M^{-1}K \\ -K^\top M^{-1} & \Sigma_{BB}^{-1} + K^\top M^{-1} K \end{pmatrix}.$$

Using $\Sigma_{AA} - K\Sigma_{BA} = M$ and $\Sigma_{BB}K^\top = \Sigma_{BA}$, the blocks of $\Sigma\Omega'$ are

$$\begin{aligned} \Sigma_{AA}M^{-1} - \Sigma_{AB}K^\top M^{-1} &= MM^{-1} = I, \\ -\Sigma_{AA}M^{-1}K + \Sigma_{AB}\Sigma_{BB}^{-1} + \Sigma_{AB}K^\top M^{-1}K &= -MM^{-1}K + K = 0, \\ \Sigma_{BA}M^{-1} - \Sigma_{BB}K^\top M^{-1} &= 0, \\ -\Sigma_{BA}M^{-1}K + I + \Sigma_{BB}K^\top M^{-1}K &= I. \end{aligned}$$

So $\Omega = \Omega'$: $\Omega_{AA} = M^{-1}$, $\Omega_{AB} = -M^{-1}K = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.

(b)

Step 3. $R = X_A - \mu_A - K(X_B - \mu_B)$ has $\mathrm{Cov}(R, X_B) = \Sigma_{AB} - K\Sigma_{BB} = 0$ and

$$\mathrm{Var}(R) = \Sigma_{AA} - K\Sigma_{BA} - \Sigma_{AB}K^\top + K\Sigma_{BB}K^\top = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = M.$$

Step 4. (R, X_B) is jointly Gaussian and uncorrelated, hence independent: $R \sim N(0, M) \perp\!\!\!\perp X_B$, and $X_A = \mu_A + K(x_B - \mu_B) + R$ given $X_B = x_B$. $M = (\Omega_{AA})^{-1}$ is free of x_B. ■

Exercise 8.2 (When the graphical lasso returns no edge)

Let $S_{jj} > 0$ for all j and $\lambda \geq \max_{j \neq k} |S_{jk}|$, $\lambda > 0$. Show that $\widehat{\Omega}_\lambda = \mathrm{diag}(1/S_{11}, \ldots, 1/S_{pp})$, and that this minimizer is unique.

Solution [Supplement].

Step 1. (Strict convexity.) For $\Omega \succ 0$, symmetric $\Delta \neq 0$,

$$\frac{d^2}{dt^2}\{-\log\det(\Omega + t\Delta)\}\Big|_{t=0} = \mathrm{tr}(\Omega^{-1}\Delta\Omega^{-1}\Delta) = \|\Omega^{-1/2}\Delta\Omega^{-1/2}\|_F^2 > 0,$$

so $f(\Omega) = \mathrm{tr}(S\Omega) - \log\det\Omega + \lambda\sum_{j\neq k}|\Omega_{jk}|$ is strictly convex; a point with $0 \in \partial f$ is the unique minimizer.

Step 2. (Optimality.) $\nabla[\mathrm{tr}(S\Omega) - \log\det\Omega] = S - \Omega^{-1}$, so $0 \in \partial f(\Omega) \iff \Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jj} = 0$, $\Gamma_{jk} = \mathrm{sign}(\Omega_{jk})$ if $\Omega_{jk} \neq 0$, $\Gamma_{jk} \in [-1, 1]$ otherwise.

Step 3. (Check.) $\Omega_0 = \mathrm{diag}(1/S_{jj})$: diagonal $S_{jj} - S_{jj} = 0$; off-diagonal $\Gamma_{jk} = -S_{jk}/\lambda \in [-1, 1]$ gives $0 - S_{jk} - \lambda\Gamma_{jk} = 0$.

Thus $\lambda_{\max} = \max_{j \neq k} |S_{jk}|$ starts the path (the largest absolute correlation for standardized data). ■

Exercise 8.3 (Partial correlation from the two node-wise regressions)

With $\beta_k^{(j)}$ the population coefficient of X_k in the regression of X_j on X_{-j}, show that for $j \neq k$

$$\rho^2_{jk\cdot -\{j,k\}} = \beta_k^{(j)} \beta_j^{(k)}, \qquad \beta_k^{(j)} = \rho_{jk\cdot -\{j,k\}} \sqrt{\frac{\mathrm{Var}(X_j \mid X_{-j})}{\mathrm{Var}(X_k \mid X_{-k})}},$$

so $\beta_k^{(j)}$, $\beta_j^{(k)}$ and $\rho_{jk\cdot -\{j,k\}}$ have the same sign and vanish together. Conclude that the AND and OR rules agree in the population.

Solution [Supplement].

Step 1. $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, $\beta_j^{(k)} = -\Omega_{kj}/\Omega_{kk}$, with $\Omega_{jj}, \Omega_{kk} > 0$.

Step 2. Product $= \Omega_{jk}^2/(\Omega_{jj}\Omega_{kk}) = \rho^2_{jk\cdot -\{j,k\}}$.

Step 3. $-\Omega_{jk} = \rho_{jk\cdot -\{j,k\}} \sqrt{\Omega_{jj}\Omega_{kk}}$ gives $\beta_k^{(j)} = \rho_{jk\cdot -\{j,k\}} \sqrt{\Omega_{kk}/\Omega_{jj}}$, and $1/\Omega_{jj} = \mathrm{Var}(X_j \mid X_{-j})$.

In samples AND/OR differ only through estimation error. ■